Customer Shopping Behavior Analysis

Analyzing 3,900+ transactions to uncover revenue drivers and customer insights

Business Problem

- **Customer Visibility** — Businesses lack visibility into customer purchasing behavior
- **Discount Strategy** — Ineffective discount strategies reduce profitability
- **High-Value Customers** — Difficulty identifying high-value customers
- **Product & Retention Insights** — Limited insights into product performance and customer retention

How can data help improve revenue, customer engagement, and decision-making?

Objective

01

Identify high-value customers

02

Analyze impact of discounts on spending

03

Evaluate product and category performance

04

Provide actionable business insights

Dataset Overview

3,900

Purchase Records

Transactional data analyzed

18

Data Columns

Customer and purchase details

37

Missing Values

Review ratings imputed

Key Features

- Demographics: Age, Gender, Location
- Purchase details: Category, Amount, Season
- Behavior: Discounts, Reviews, Shipping

Analysis Tools

Python for data cleaning and feature engineering, SQL Server for structured queries, Power BI for visualization

Revenue by Gender

Male Customers

$157,890 total revenue

Higher spending segment

Female Customers

$75,191 total revenue

Significant market share

Product Performance Insights

Top-Rated Products

1. Gloves (3.86 avg)
2. Sandals (3.84 avg)
3. Boots (3.82 avg)
4. Hat (3.80 avg)
5. Skirt (3.78 avg)

Most Discounted Items

1. Hat (50% discounted)
2. Sneakers (49.66%)
3. Coat (49.07%)
4. Sweater (48.17%)
5. Pants (47.37%)

Shipping & Subscription Impact

Express Shipping

$60.48 average purchase

Premium shipping drives higher spend

Standard Shipping

$58.46 average purchase

Standard option for budget-conscious

Non-Subscribers

$59.87 avg spend, 2,847 customers

Majority segment with higher revenue

Subscribers

$59.49 avg spend, 1,053 customers

Lower average but loyal base

Customer Segmentation

01

Loyal Customers

3,116 customers with consistent purchases

02

Returning Customers

701 customers making repeat visits

03

New Customers

83 customers in initial purchase phase

Top Products by Category

Accessories

- Jewelry (171 orders)
- Sunglasses (161)
- Belt (161)

Clothing

- Blouse (171 orders)
- Pants (171)
- Shirt (169)

Footwear

- Sandals (160 orders)
- Shoes (150)
- Sneakers (145)

Outerwear

- Jacket (163 orders)
- Coat (161)

Revenue by Age Group

Young Adult

$62,143 revenue

Middle-aged

$59,197 revenue

Adult

$55,978 revenue

Senior

$55,763 revenue

Key Findings

Repeat Buyers Subscribe

958 repeat buyers subscribed vs. 2,518 non-subscribers with >5 purchases

Discount Strategy

839 customers used discounts while spending above average ($59.76)

Age Group Balance

Young adults lead revenue ($62,143), seniors close behind ($55,763)

Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers

Customer Loyalty Programs

Reward repeat buyers to move into loyal segment

Review Discount Policy

Balance sales boosts with margin control

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

Why This Matters

- Helps businesses target the right customers
- Improves marketing and discount strategies
- Increases customer retention and lifetime value
- Supports data-driven decision-making

> “Data-driven insights can directly impact revenue and business growth.”